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Rigless Well Intervention Deploying Straddle Packer System

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Abstract

This paper presents an in-depth analysis of a low-cost solution designed to economically sustain production and avoid high workover expenses. It explores the technical and commercial aspects of straddle packer technology as an alternative to traditional rig-based workovers, which typically require tubing replacement. The discussion also covers the optimum deployment method for this technology and the operational risks involved.

In gas lift wells, a key factor in production optimization is deepening the gas injection point as much as possible to lower bottomhole flowing pressure and enhance drawdown—especially in wells characterized by a high productivity index. One of the most challenging constraints to achieving this is the presence of holes in the tubing string. The straddle packer offers an effective solution, as it seals off tubing holes (regardless of length), thereby allowing gas injection at a deeper point. Success depends heavily on accurate depth correlation and precise setting of the straddle.

This paper introduces a case study involving the deployment of a straddle packer spaced with a 25 ft spacer to cover a 10 ft long crack in N-80 tubing. Three deployment methods were evaluated: slickline, electric line, and digital slickline. While slickline offered the advantage of mechanical jarring, it lacked correlation tools (GR or CCL). Electric line provided precise depth correlation but couldn't ensure proper straddle setting. The optimal solution was digital slickline, which combined built-in CCL for accurate depth correlation with jarring capability. This method was successfully used to restore a long-term shut-in well, resulting in a production gain of 600 BOPD and recovery of over 0.5 MMSTB of reserves. The well was the only sink point in the reservoir compartment.

The operation proved to be an excellent alternative to costly rig-based workovers—particularly offshore, where mobilization costs are high. Additionally, it presented significantly less risk to the pay zone (no kill fluid invasion or iron rust contamination) and reduced operational complexity.

Introduction

Gulf of Suez oil company is one the largest gas lift operators in Egypt, The company has operated more than 600 oil wells that has been gas lifted throughout the entire company history with more than 500 MMSCFPD

are being injected daily. Gas lift systems comprise various components such as gas injection mandrels, lift valves, and tubing. The design of these elements to function effectively in unity, while also enduring high pressures and corrosive conditions, presents significant challenges. Precise engineering and controls are required to achieve successful systems in which a diverse range of operating conditions, including varying pressures, temperatures, and fluid properties, can be handled. It is also essential to regulate the volume of the gas introduced into the well since insufficient gas may fail to generate adequate lift, whereas excessive gas can result in gas locking or suboptimal performance. Although the gas lift can be considered one of the most efficient and reliable artificial lift techniques, it usually suffers from leak points across the tubing body (including the side-pocket mandrels) that affects the well performance.

Case Study Background

The well discussed in this case study is located in the October field that had started production from Cretaceous sandstone formation with stable liquid production of ~ 1600 BLPD on June 94 from N-80 4.5×3.5" completion string. The well ceased to produce in 2014. The sudden drop in well productivity is mainly attributed to a mechanical problem(s) as shown in Fig 1.

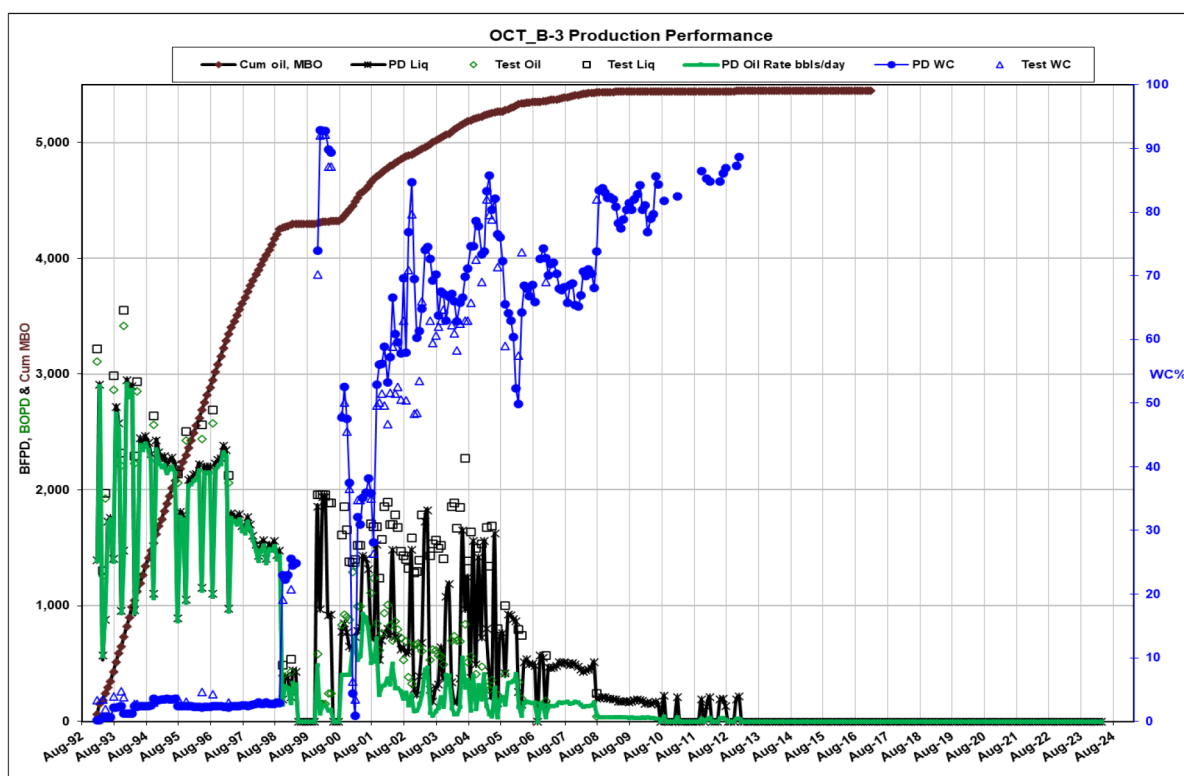


Figure 1—Production Performance of the Well

The well had produced a total of 5.5 MMSTB. The remaining reserve is estimated to be of 0.5 MMSTB in this compartment. This can not be drained by any other wells (the well location considered as solo producer in this area). In order to access the remaining reserve several diagnostic logs had been run in the well to identify the suspected mechanical issue.

In addition to the impact on well production and reserve loss, another crucial point is the well completion integrity. The integrity of well completion plays a significant role in maintaining the integrity of the well as a whole. If there are issues with the completion, it can lead to problems with the wellbore, such as leaks, sand production, and causing failures, which can compromise the safety and productivity of the well.

Therefore, it is essential to ensure that the completion design, installation, and testing are carried out to high standards to prevent any issues that may arise during the production life of the well. Regular inspection, testing, and maintenance of the completion components are also necessary to ensure their continued functionality and detect any early signs of potential failure.

Periodic tubing and annulus communication test shall be performed for natural flow wells every two years (24 months), after GLVC or abrupt changes in gas lift DS pressure. The last test performed for the subject well indicated severe communication. Accordingly, a complete diagnostic plan has been set in order to precisely determine the communication location and the gas rate passing through it.

Diagnostic logs

The first log that had been run in the subject well was the multi-finger caliper log that showed a hole at joint# 207 in the 3.5" tubing string which located at depth 6456 ft THF as illustrated in the Figure below.

The following log shows a 100% penetration that confirmed the presence of clear hole while the maximum penetration in the other joints was 30%. Based on the log, there is a clear hole in the tubing string and not in a gas lift valve or side pocket mandrel.

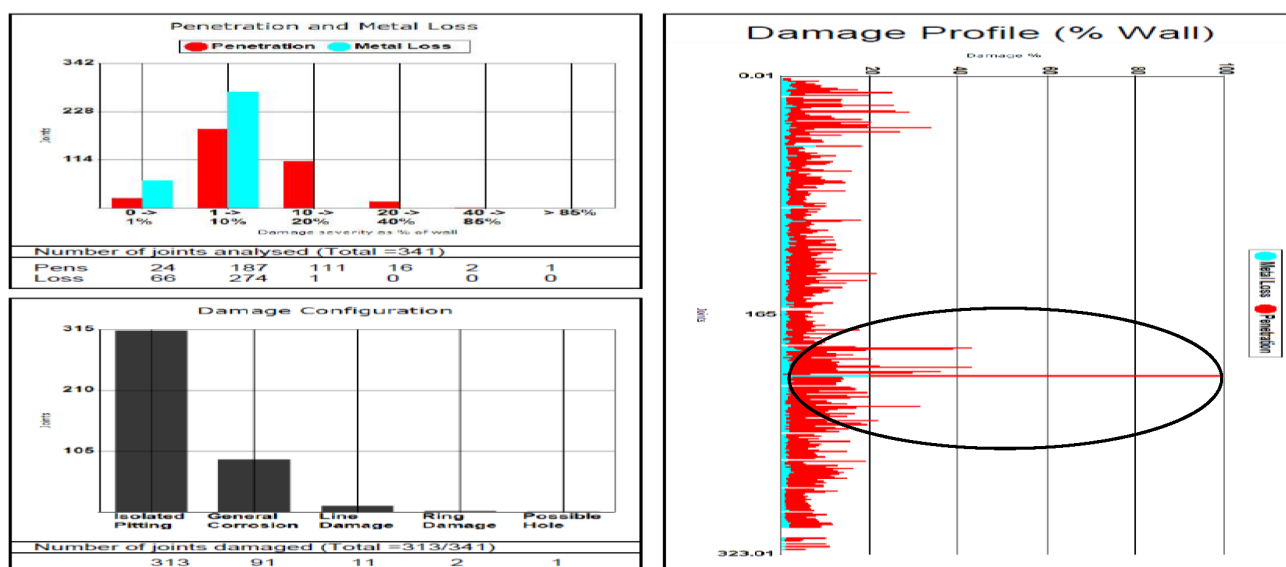


Figure 2—MFCL Results in the Well

The second log that had been run is a continuous P&T survey for precise determination of the hole location in addition to confirm the depth of the hole. while applying high pressure gas from the tubing side to the annulus side (i.e. reverse gas lift injection) so that any leak point would be clearly identified with the temperature log that very sensitive to any induced flow especially that the gas lift injected from the GL modules had lower temperature than the ambient temperature (i.e. clear cooling effect will appear against leak point)

The log showed the leak point at [figure 3](#) at 6513 ft ORKB after correlation with the tubing accessories. This log in addition to confirm that there is no other leak points it was also a key player in determining the exact depth of the wash pipe that will be directly against the pipe hole.

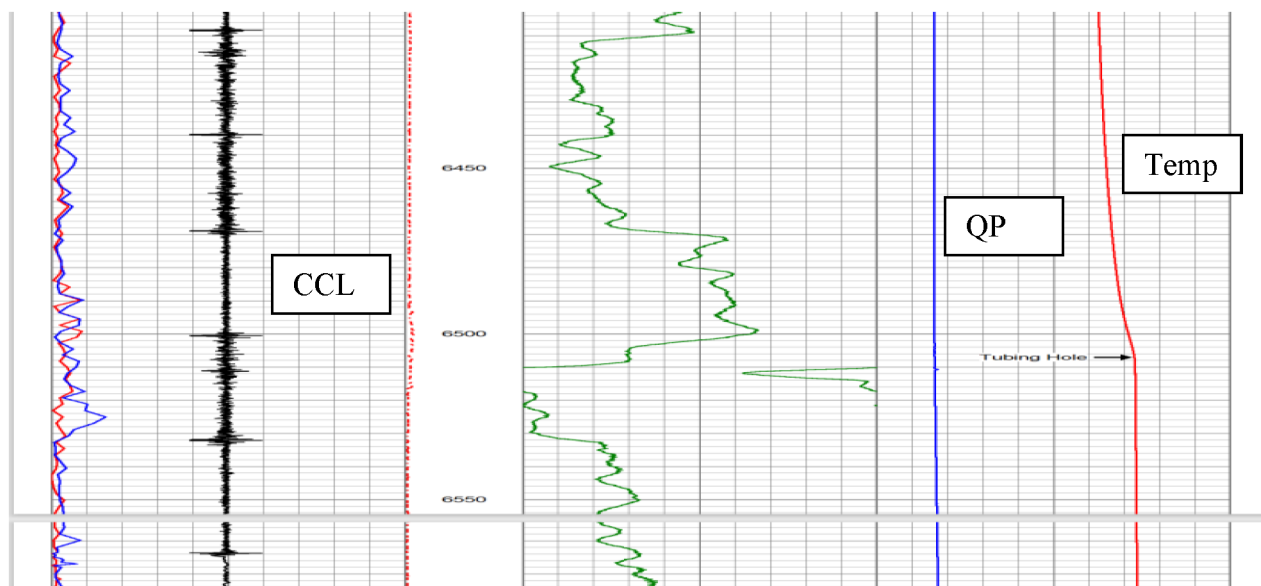


Figure 3—Pressure and Temperature Log Across the Hole

After confirming the leak depth nodal analysis simulation run showed that the fluid level is just below the communication point depth which deprives the injected to create any draw-down and this the main reason that the well is not producing any liquid and just circulating the gas lift

On the other hand, if we succeeded lowering the point of gas injection till it reached the maximum allowable depth then well would be capable of producing ~ 1200 BLPD with 300 BOPD oil gain as clarified in fig.4

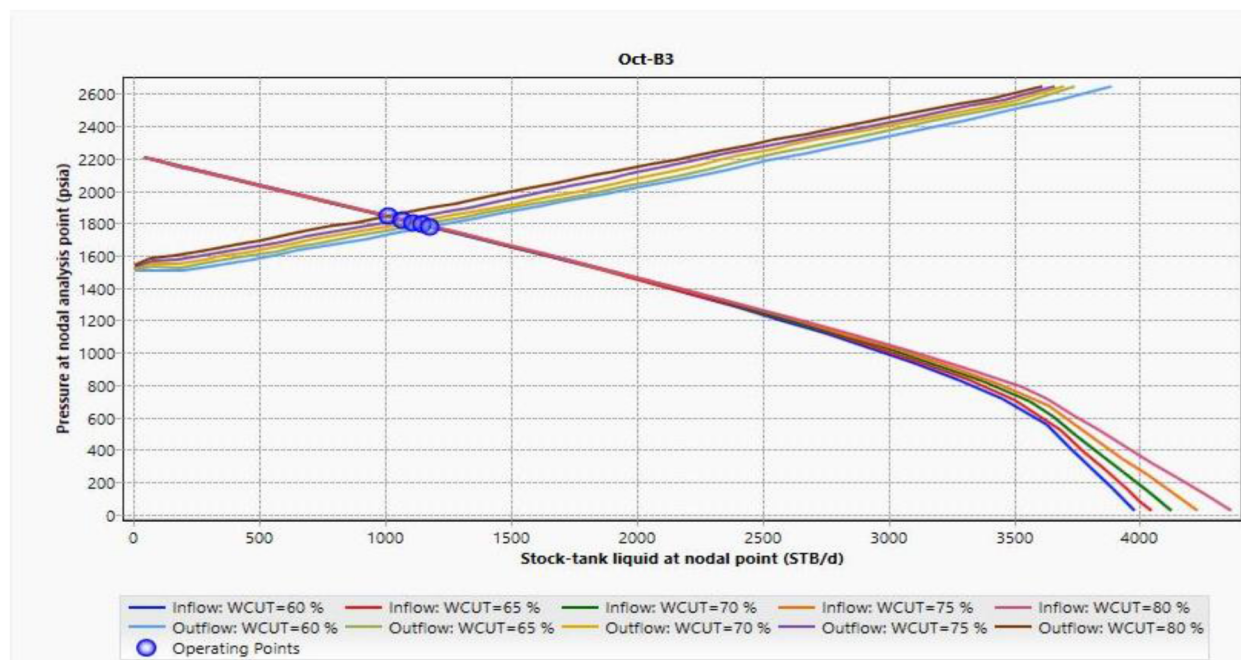


Figure 4—Nodal Analysis Simulation Run

Methodology

Straddle Packer System Theory

The straddle packer system operates on the principle of creating a hydraulic bypass around damaged tubing sections while maintaining integrity and production capability. The system consists of two mechanical packers connected by a production spacer that enables fluid flow to bypass the compromised tubing section. The implemented methodology provides significant cost efficiency by eliminating the requirement for completion retrieval, reducing offshore rig time requirements, and minimizing operational risk in the offshore environment. Technical benefits include maintaining production capability during remediation, providing pressure integrity up to system ratings, and enabling reversible intervention without permanent completion modification.

Key Definitions

Retrievable Straddle Packer is defined as a dual-packer assembly designed for temporary isolation of tubing sections, consisting of upper and lower sealing elements connected by a flow spacer. Digital Slickline refers to an electric wireline system providing real-time depth correlation and tool activation capabilities. The Anchor Seal Assembly represents a mechanical connection system enabling staged installation of straddle packer components. Tubing Leak Isolation describes the process of hydraulically bypassing damaged tubing sections while maintaining production flow path.

System Design

The straddle packer configuration creates a hydraulic bypass around the tubing leak through a specifically engineered flow path. Production fluids enter the system below the lower packer and are directed through the lower packer into the spacer. Fluids continue through the spacer, bypassing the leak zone, before exiting through the upper packer into the existing tubing above the leak. This design maintains well productivity while isolating the compromised tubing section, eliminating the need for costly completion retrieval operations in the offshore environment.

Description and Application of Equipment and Processes

The straddle packer system was selected based on wellbore constraints and operational requirements for offshore applications. The existing tubing configuration consisted of 3½ inch tubing with 2.992 inches internal diameter containing a leak at 6513 ft ORKB. The straddle packer specifications included a maximum outside diameter of 2.72 inches to ensure passage through the 2.992 inch ID tubing, with a production bore of 1.81 inches ID post installation. The system was rated for 5,000 psi pressure and 275°F temperature operation, utilizing a retrievable mechanical setting mechanism.

Deployment equipment comprised a digital slickline unit with real-time depth correlation capabilities, gauge cutter for wellbore conditioning, and packer setting tool with timed activation capability. The equipment selection prioritized operational efficiency while maintaining system reliability under offshore conditions.

Operational Methodology

The methodology was developed to address offshore operational constraints, specifically eliminating the need for completion retrieval and minimizing rig time requirements. The approach consists of three distinct operational phases designed to ensure successful straddle packer installation and system activation.

Phase 1: Wellbore Assessment and Preparation. The first phase focused on verifying wellbore accessibility and establishing operational parameters critical for successful packer deployment. A gauge cutter was deployed on digital slickline to the target depth to condition the wellbore and verify unrestricted access to the setting depth. Baseline conveyance time was established for accurate depth correlation, while

wellbore conditions were recorded to confirm setting depth accessibility. The packer setting tool activation timing was calibrated based on conveyance data to ensure precise operational control.

Phase 2: Lower Packer Installation. The second phase involved installing and setting the lower packer element below the leak zone. The lower packer assembly was deployed on digital slickline utilizing real-time depth correlation for precise positioning below the identified leak zone. Following successful positioning, the lower packer was set and seal integrity verified through pressure testing. This phase established the anchor point for subsequent upper assembly installation, ensuring system stability and proper load distribution.

Phase 3: Upper Assembly Installation and System Activation. The final phase completed straddle packer installation and established the bypass flow path. The upper packer assembly with spacer and anchor seal was deployed on digital slickline, engaging the anchor seal mechanism with the previously set lower packer. The upper packer was positioned above the leak zone and set to complete the isolation system. A continuous flow path was established through the 1.81-inch ID production bore, followed by comprehensive system pressure testing to verify operational integrity.

Operational Procedures

Pre-Operational Technical Requirements

Well Preparation Standards. The operational sequence requires specific well preparation protocols to ensure safe and successful straddle packer installation. The well must remain shut-in for a minimum of 24 hours before operation commencement to stabilize wellbore conditions and establish proper safety protocols. A comprehensive clean-out run using high-pressure jetting tools is mandatory to ensure the packer setting depth remains clear of debris and operational obstructions. The drift and clean-out run serves as verification that the target packer setting depth is accessible and free from restrictions that could compromise tool deployment or setting operations.

In cases where hard scale presence is anticipated based on well history or formation characteristics, a specialized milling bottom hole assembly is recommended for scale removal prior to packer installation operations. Well control responsibilities are clearly defined and shared between the slickline service provider and the customer to maintain adequate barrier systems throughout all operational phases.

Slickline Service Provider Requirements. Critical slickline equipment specifications must be verified and confirmed prior to operational commencement. The available lubricator length must accommodate the maximum running bottom hole assembly configuration for each of the three planned operational phases. Slickline wire specifications are determined by the wireline team based on specific operational requirements, wellbore conditions, and anticipated load requirements during installation procedures.

Maximum pull and push capabilities at the target setting depth must be confirmed to ensure adequate tool manipulation during installation sequences. Slickline grade and specifications must meet operational load requirements throughout the installation process. Jarring capability is essential for engaging the anchor seal assembly with the previously installed lower packer, while up-jarring capability using power jars or spring jars is required for contingency packer release operations if retrieval becomes necessary.

Mechanical and electronic depth measurement systems require recent calibration to prevent depth correlation errors during precise packer positioning operations. Accurate depth control is critical for proper packer placement relative to the tubing leak location and successful anchor seal engagement between upper and lower packer assemblies.

Standard Operating Procedures

Packer to Setting Tool Assembly. The packer assembly to the Setting Tool follows specific sequential procedures to ensure proper tool configuration and operational readiness. The correct quantity of shear screws must be confirmed and installed in the release collet assembly according to manufacturer specifications. The thrust sleeve is carefully positioned over the setting tool, ensuring proper alignment and clearance for operational functions.

The adapter sub is made up to the setting tool using appropriate torque values, with the linear actuator housing serving as backup during installation to prevent over-torquing or component damage. Set screws are installed and torqued to specification to secure all connections. The inner mandrel is connected to the slick rod of the setting tool using proper torque application, with the linear actuator housing maintained as backup during this critical connection.

The packer is assembled to the release collet using hand-tight connections with strap wrenches to prevent over-torquing that could damage sealing elements. The thrust sleeve is threaded onto the adapter sub with precise positioning, maintaining a quarter-inch gap between the thrust sleeve and the top of the WidePak packer to ensure proper operational clearance.

Three-Phase Operational Sequence

First Run: Dummy Run - Wellbore Assessment. The dummy run utilizes a bottom hole assembly consisting of a 2.85-inch gauge cutter designed to verify wellbore accessibility and establish operational parameters. The assembly is made up inside the lubricator and tested according to operational requirements before deployment. The tool is run into the hole at 75 feet per minute to 6,510 feet depth, with time recording maintained to establish optimal IEST timer settings for subsequent packer installation operations.

The E-line engineer correlates depth measurements and establishes the most effective setting depth for packer placement relative to the tubing leak location at 6,467 feet. Following successful completion of wellbore assessment, the tool is retrieved to surface at 75 feet per minute, providing essential baseline data for the subsequent packer installation phases.

Second Run: Lower Packer Installation. The lower packer installation phase utilizes a packer system with a setting tool provided with a timer for precise setting control. The bottom hole assembly is made up with a 25-foot extension as planned, with assembly installation inside the lubricator and testing performed according to operational requirements. The tool string is run into the hole at 75 feet per minute to the target setting depth of 6,470-6,475 feet, maintaining precise depth correlation with previous run data and taking weight measurements every 1,000 feet during descent.

At the designated setting depth, the assembly is brought to static weight and brakes are applied to maintain position. The setting tool timer is allowed to elapse, initiating the packer setting sequence. Sufficient time is allocated for complete packer setting based on wellbore conditions and manufacturer recommendations. Once the packer is successfully set, the IEST tool is released from the packer assembly and retrieved at 20 feet per minute, with careful monitoring for overpull conditions that could indicate incomplete release or complications.

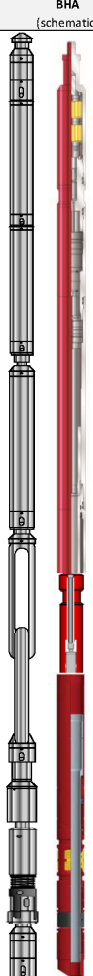
Third Run: Upper Packer with Spacer and Anchor Seal Installation. The final operational phase involves installation of the upper packer assembly with integrated spacer and anchor seal system. The bottom hole assembly is made up and installed inside the lubricator with comprehensive testing performed according to operational protocols. The assembly is run into the hole at 75 feet per minute to the target depth above the previously installed lower packer.

At 50 feet above the setting depth, weight was checked record run-in and pick-up weights for operational correlation and verification purposes. The assembly is then run slowly at 10 feet per minute to 6,470 feet depth to engage the anchor seal mechanism with the previously set lower packer. Precise weight control is

maintained during this critical engagement phase to ensure proper mechanical connection between upper and lower packer assemblies.

A controlled pick-up test applying 500 pounds overpull is performed to verify proper anchor seal latching, with this overpull maintained for 2 minutes to confirm secure engagement. The assembly is then set to neutral weight, and the timer activation sequence is initiated. Sufficient time is allowed for complete packer setting before retrieval operations commence at 75 feet per minute, completing the straddle packer installation and establishing the production bypass system.

Bottom Hole Assembly

BHA (schematic)	Item	Description	OD (in.)	ID (in.)	Length (ft.)	Connection	Fish Neck (In)	Serial Number
	1	<u>SLICKLINE ROPE SOCKET</u>	1.875	N/A	1	1.875" QLS (BOX)	1.75	
	2	<u>STEM BAR</u>	1.875	N/A	5	1.875" QLS (PIN x BOX)	1.75	
	3	<u>STEM BAR</u>	1.875	N/A	5	1.875" QLS (PIN x BOX)	1.75	
	4	<u>SPANG JAR</u>	1.875	N/A	7.25	1.875" QLS (PIN x BOX)	1.75	
	5	<u>CROSSOVER</u>	1.875	N/A	0.46	1.875" QLS (PIN) X 0.9375" -10UN BOX	1.75	
	6	<u>2.5" NOM JDC/SB PULLING TOOL</u>	2.235	N/A	1.2	0.9375"-10 UN PIN	1.375	
	7	<u>SLICKLINE DUMMY ROPE SOCKET</u>	1.875	NA	1	1.060-in. 10 TPI UN BOX	1.75	
	8	<u>ISO EASYSET SETTING TOOL</u> WEIGHT: 38 lbs MAXIMUM STROKE : 10" PRESSURE RATING: 15,000 psi TEMPERATURE RATING: 32 Deg F - 275 DegF NOM OPERATING SETTING FORCE: 30,000 lbs MAX GENERATED FORCE: 40,000 lbs	2.125	N/A	4.2	<u>UP THREAD</u> 1.060-in. 10 TPI UN PIN <u>OUTER THREAD DOWN</u> 2.000-in. 10 TPI STUB ACME PIN <u>INNER TREAD DOWN</u> 11/16 (0.688)-in. 16 TPI UN BOX	1.74	
	9	<u>WIRELINE ADAPTER KIT-27 SIZE</u> SETTING FORCE - 12,300 LBS (2,050 LBS PER SCREW X 6 EA)	2.7	NA	1.66	0.687-16 UN PIN X 2" - 10 STUB ACME BOX	2.7	
	10	<u>3.5" WIDEPACK PACKER II</u> TUBING SIZE: 3-1/2in TUBING WEIGHT: 9.2-10.2 lbs/ft MAX GAUGE RING: 2.720 in UPPER SEAL BORE: 2.187 in TEMP RATING: 275 DEG F PRESSURE RATING : 5000 PSI RELEASE FORCE: 3,700 lbs SETTING FORCE: 12,300 lbs	2.72	1.81	5.2	2 3/8" NU 10 RD PIN DOWN	2.7" W/P RETRIEVAL TOOL	

Maximum O/D	➡	2.72 in	Overall BHA Length (ft) ➡	31.97
Minimum I/D	➡	N/A		

Figure 5—Lower Packer Assembly

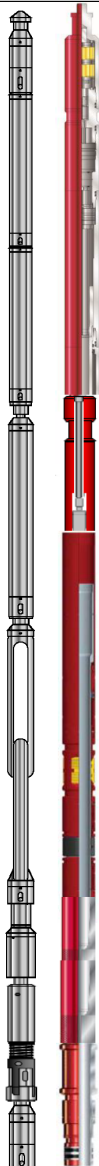
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	3	<u>STEM BAR</u>	1.875	N/A	3	1.875" QLS (PIN x BOX)	1.75										
	4	<u>SPANG JAR</u>	1.875	N/A	7.25	1.875" QLS (PIN x BOX)	1.75										
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	14	<u>SPACER PIPE</u>	2.72	1.7	15	2.0600" 20 NS-3 PIN X 2 3/8" NU 10 RD PIN	2.72										
	15	<u>WP SEAL ASSEMBLY</u> SHEAR VALUE PER SCREW: 925 lbs MIN SHEAR RELEASE: 1,850 lbs MAX SHEAR RELEASE: 11,100 lbs TEMPERATURE RATING: 75 - 275 F MAX NO: OF SHEAR SCREWS: 12	2.72	1.75	2.45	2 3/8" NU 10 RD BOX UP	2.74										
	<table><tr><td>Maximum O/D</td><td>➡</td><td>2.72 in</td><td rowspan="2">Overall BHA Length (ft) ➡</td><td rowspan="2">47.42</td></tr><tr><td>Minimum I/D</td><td>➡</td><td>N/A</td></tr></table>									Maximum O/D	➡	2.72 in	Overall BHA Length (ft) ➡	47.42	Minimum I/D	➡	N/A
	Maximum O/D	➡	2.72 in	Overall BHA Length (ft) ➡	47.42												
	Minimum I/D	➡	N/A														

Figure 6—Upper Packer Assembly

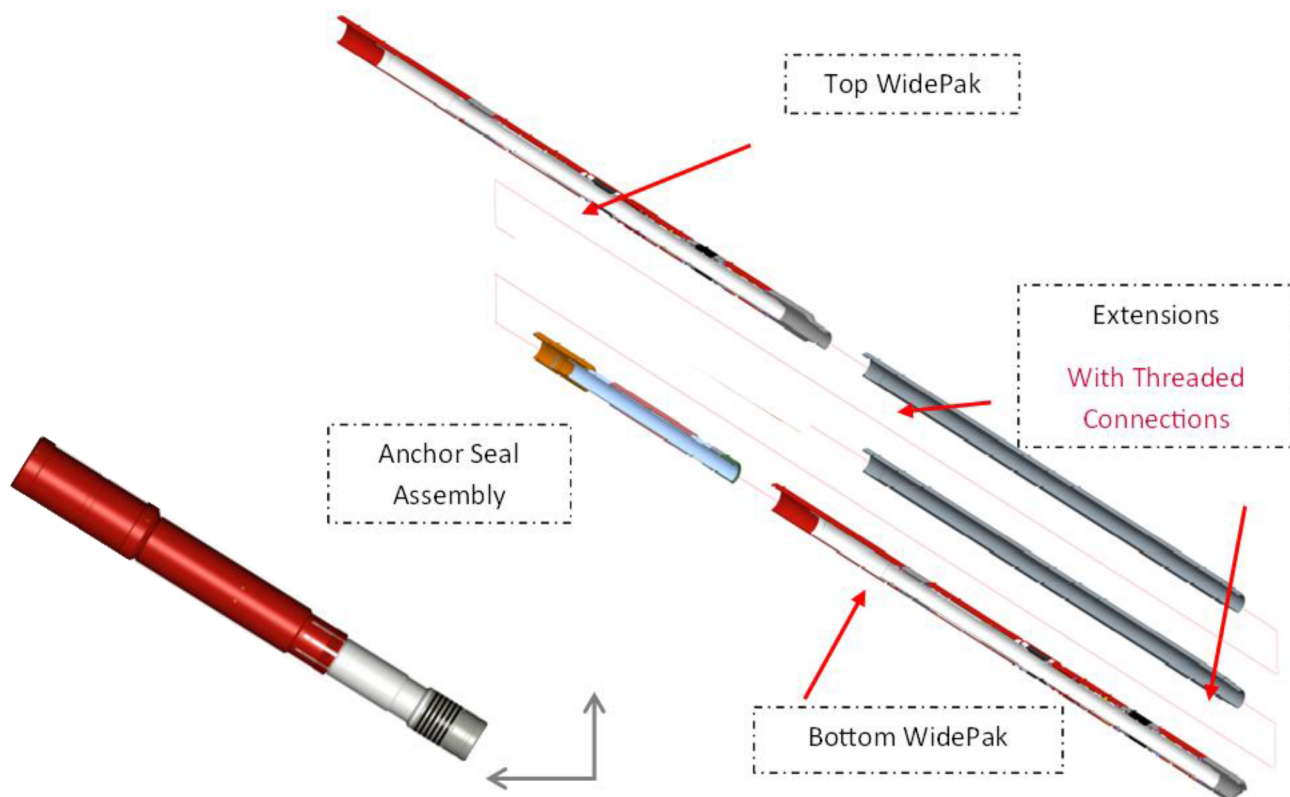


Figure 7—Straddle Packer Layout

Conclusion

The successful deployment of the straddle packer system in this case study demonstrates a revolution shift in offshore well intervention strategies, offering a technically sound and economically viable alternative to conventional rig-based workovers. The implementation of digital slickline technology proved instrumental in achieving precise depth correlation and reliable tool deployment, combining the mechanical advantages of jarring capability with the accuracy of built-in CCL systems.

The technical achievements of this operation are significant. The restoration of a long-term shut-in well resulted in immediate production gains of 300 BOPD and enabled the recovery of over 0.5 MMSTB of previously inaccessible reserves from a sole-producer reservoir compartment. The straddle packer system effectively bypassed a 10-foot tubing crack at 6,456 ft THT, demonstrating the technology's capability to address substantial mechanical failures without compromising well integrity or production capacity.

From an economic perspective, the intervention avoided the substantial costs associated with offshore rig mobilization, completion retrieval, and tubing replacement operations. The rigless approach eliminated the need for kill fluid invasion, preventing formation damage and iron rust contamination that could compromise the pay zone. This methodology significantly reduced operational complexity while maintaining full production capability throughout the remediation process.

The operational methodology developed through this case study establishes a scalable framework for similar interventions across gas lift operations. The three-phase deployment sequence wellbore assessment, lower packer installation, and upper assembly completion provides a systematic approach that minimizes operational risk while ensuring reliable system installation. The integration of digital slickline technology with mechanical setting tools offers optimal balance between precision and operational flexibility.

The implications for future offshore operations are considerable. This technology enables operators to economically sustain production in aging assets where traditional workover methods would be prohibitively

expensive or technically challenging. The ability to isolate tubing leaks regardless of their length or location provides operators with a valuable tool for extending well life and maximizing reserve recovery.

The success of this intervention validates straddle packer technology as a mature, reliable solution for addressing tubing integrity issues in gas lift wells. The methodology presented can be readily adapted to various wellbore configurations and operational environments, offering the oil and gas industry a cost-effective approach to well intervention that maintains production continuity while addressing critical mechanical failures.

For operators managing extensive gas lift systems, particularly in offshore environments, this technology represents a strategic capability that can significantly enhance the asset economics and optimize oil production. The demonstrated ability to restore wells previously considered uneconomic for conventional workover operations opens new possibilities for reserve recovery and field development strategies.

Abbreviations

GR	– Gamma Ray
CCL	– Casing Collar Locator
QP	– Quartz pressure
GLVC	– Gas Lift Valve Change
DS Pressure	– Downstream Pressure
ORKB	– Original Rotary Kelly Bushing
THT	– Tubing Hanger Tieback